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REVIEW

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Advancements in artificial intelligence technology for improving animal welfare: Current applications and research progress

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Abstract

The integration of Artificial Intelligence (AI) in various sectors has led to significant advancements, with the animal industry being no exception. This review aims to investigate the benefits, limitations, and future prospects of AI technology in improving animal welfare. First, it examines the role of AI in understanding animal behaviors and emotions, providing deeper insights into their well-being and sources of stress. Next, the paper explores how AI can revolutionize animal nutrition through innovative algorithms and data analytics. The health aspect emphasizes the ability of AI to identify and manage illnesses through intelligent systems. This review also highlights the application of AI in improving animal living conditions, with a focus on environmental management and automated cleaning and disinfection systems. In conclusion, the review emphasizes AI-driven techniques for early prediction, close monitoring, and accurate diagnosis of animal diseases, ensuring healthier and more sustainable livestock management. By leveraging its advantages, addressing limitations, and exploring future directions, AI has the potential to significantly enhance animal welfare, sustainable agriculture, and veterinary practices.

Keywords

animal behavior recognition, animal emotion recognition, animal welfare, artificial intelligence, intelligent health systems

INTRODUCTION

Animals play a crucial role in our ecosystem and coexist with us, forming an intricate web of life [1]. As sentient beings, they deserve not only respect but also our vigilant attention to their welfare. The importance of animal welfare has gained widespread recognition in recent years as we grapple with the ethical and moral obligations associated with our treatment of animals [1–3]. Whether they are farm animals, companion pets, wildlife, or laboratory subjects, ensuring their welfare has become an imperative in our modern times [4, 5].

Concurrently, the field of Artificial Intelligence (AI) has witnessed unprecedented advancements, encompassing machine learning, deep learning, and neural

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networks, demonstrating remarkable potential in numerous domains [6, 7]. One of the most promising and ethically significant applications of AI is its integration into the realm of animal welfare [8]. By leveraging AI technologies, we have the opportunity to revolutionize the monitoring, management, and enhancement of animal lives across various contexts [9].

This comprehensive review aims to delve into the synergistic relationship between AI and animal welfare, shedding light on the multifaceted applications of AI in this field. Specifically, we explore how AI is employed to address key aspects of animal welfare, including nutrition, habitat improvement, health care, and emotional recognition (refer to Table 1 and Figure 1). By elucidating these applications, we intend to emphasize the profound significance of AI in reshaping our approach to animal welfare and fostering a more compassionate and informed coexistence between humans and animals. Through this exploration, we aspire to inspire further research and innovation, ultimately advancing the cause of animal welfare in our ever-changing world.

THE IMPACT OF AI TECHNOLOGY ON ANIMAL BEHAVIOR AND EMOTION RECOGNITION

Animal behavior recognition

AI technology is revolutionizing our ability to recognize and understand animal behavior, offering numerous benefits for animal welfare, conservation, research, and farm management [34]. Traditional methods of visual recognition have limitations, but AI-powered solutions enhance accuracy and efficiency in identifying individual animals and monitoring their behavior [35]. These advancements hold great promise for improving animal well-being and the management of animal populations [36].

While visual recognition has been widely used for animal identification, it often requires human assistance and has limitations in terms of accuracy and long-term recognition [37, 38]. Recently, there has been a focus on improving recognition accuracy and efficiency through the introduction of AI. Efficient individual recognition can help quickly obtain individual information to ensure animal welfare. In a recent study, deep learning techniques were applied to develop a novel face recognition model based on convolutional neural networks (CNNs) for identifying individual giant pandas [10]. The model successfully recognized 95% of the giant panda individuals in the validation dataset. Even under simulated field conditions with decreased photo data quality, the model still accurately identified over 90% of giant panda individuals. By leveraging deep learning and advanced recognition models, researchers can enhance the accuracy and efficiency of individual identification in animals [10]. This technology holds promise for various applications in conservation, research, and wildlife management, ultimately contributing to the well-being and conservation of animal populations.

In addition to visual algorithms, collecting relevant data on animal behavior through sensors and training AI algorithms based on that data is also commonly used to improve animal welfare and farm management efficiency [39]. B. Ji et al. developed a machine learning framework by collecting behavior, health, and productivity data from 80 cows on a dairy farm over a period of 5 years (Figure 2A.) [11]. They predicted daily milk production, composition (fat and protein content), and milking frequency for each cow over the subsequent 28 days. The experimental results demonstrated an accuracy rate of over 80% for this learning framework. The authors aim to integrate this framework with robotic milking systems to determine healthier milking strategies for cows. By leveraging sensor data and machine learning, researchers can gain valuable insights into animal behavior and well-being, leading to improved management practices and enhanced animal welfare [39]. These technological advancements offer promising possibilities for optimizing various aspects of animal care and husbandry.

In challenging environments, accurately identifying animal behavior using visual recognition and activity sensors can be difficult [40]. However, Manteuffel addressed this issue by employing radar sensors, which proved effective in recognizing parturition-related activities in sows (Figure 2B) [12]. By analyzing the data from 15 sows and utilizing techniques such as daily activity differences and accumulation, a simple classifier was implemented to detect deviations in activity levels. The classifier exhibited a sensitivity of 92%, a specificity of 80%, and an area under the receiver operating characteristic curve of 95%. This novel approach achieved the classification and identification of animals in different states through the analysis of daily activity level differences.

In the field of livestock farming, future poultry production could potentially optimize operations through the use of intelligent autonomous systems, remotely managed by human workers and intervening only when necessary [41]. Park et al. proposed a system that effectively ensures the welfare of poultry animals [13]. This system involves sensors and robots for round-the-clock management of the poultry house. A multifunctional robot performs most of the daily tasks, such as removing dead chickens and monitoring flock behavior to ensure the growth and welfare of the chickens.

The livestock density is a crucial factor that affects large-scale animal production and animal welfare [42]. However, the current artificial counting methods used in the Muscovy duck farming industry are inefficient,

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TABLE 1 Specific applications of AI in improving animal welfare.

Field of application		Function	Reference
The impact of AI technology on animal behavior and emotion recognition	Animal behavior recognition	A novel face recognition model based on convolutional neural networks for recognizing individual giant pandas was developed by applying deep learning techniques.	[10]
		Predicting a cow's daily milk yield, composition, and milking frequency by utilizing sensor data and machine learning.	[11]
		Identifying farrowing-related activity in sows using radar sensors.	[12]
		Intelligent robotic system applied to round-the-clock management of poultry houses.	[13]
		Smart duck counting via AI detection of image data.	[14]
	Animal emotion recognition	Analyzing domestic shorthair cat facial image data by machine learning models for automated pain recognition in domestic cats.	[15]
		Intelligent detection of equine pain signals and diagnosis by training vision algorithms with automatic computational classifiers via machine learning models.	[16]
AI technology ensuring animal nutrition and health	AI revolution in animal nutrition	An intelligent learning model for collecting cow feed intake in automatic feeding systems to develop feeding schedules.	[17]
		An intelligent feeding system for controlling pet obesity.	[18]
	Advancing animal health through intelligent systems	Predicting learning models for cow eye temperature, milk yield, and quality.	[19]
		An intelligent diagnostic system for equine lameness.	[20]
		Intelligent systems for animal health monitoring, animal tracking, milk quality and supply chain, and feed monitoring and safety.	[9]
		A machine learning model for a noninvasive video biometric system for intelligent monitoring of cow health.	[21]
The role of AI in revolutionizing animal living environments	The revolutionary application of AI in animal living environment management	Intelligent real-time forecasts for CO_2, SO_2, NO_2, PM2.5, and PM10 levels.	[22]
		Monitoring system for temperature, carbon dioxide, and wind speed in layer coops.	[23]
		Automatic monitoring and an adjustment system for livestock and poultry farming environment based on	[24]

(Continues)

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TABLE 1 (Continued)

Field of application		Function	Reference
		WiFi, LoRa wireless communication, and IoT technologies.	
	The application of automated cleaning and disinfection systems in animal facilities	Rapid disinfection strategy for livestock farms based on remote control of intelligent robots.	[25]
		Magnetic RFID-based navigation-guided intelligent robots target the full range of disinfection strategies for large livestock farms.	[26]
Advancements in AI-driven animal disease prediction, monitoring, and diagnosis	Animal disease prediction and monitoring	A fog-centered, IoT-based smart health care support system to monitor and control swine flu virus outbreaks.	[27]
		A fog computing-assisted end-to-end animal behavioral analysis and disease monitoring platform for remote farms with limited Internet connectivity.	[28]
		An IoT-based "livecare" framework for automated monitoring of the health of cows on large dairy farms.	[29]
		An IoT-based animal social behavior sensing framework for modeling the spread of mastitis in dairy cows and inferring the risk of mastitis infection in dairy cows.	[30]
	Diagnosis of animal diseases	An automated swine welfare indicator assessment system for the diagnosis of ear and tail lesions in slaughterhouse pigs.	[31]
		Convolutional neural networks to improve the diagnostic accuracy of expert systems for fast and concise animal disease diagnosis.	[32]
		Computational models developed using penalized linear regression, random forests, gradient boosters, and neural networks for helping farmers take evidence-based interventions before anticipated stressful environmental conditions occur.	[33]
		An artificial intelligence-based contactless biometric system for automated assessment of animal welfare on farms and during transportation.	[3]

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FIGURE 1 The application of AI in animal welfare.



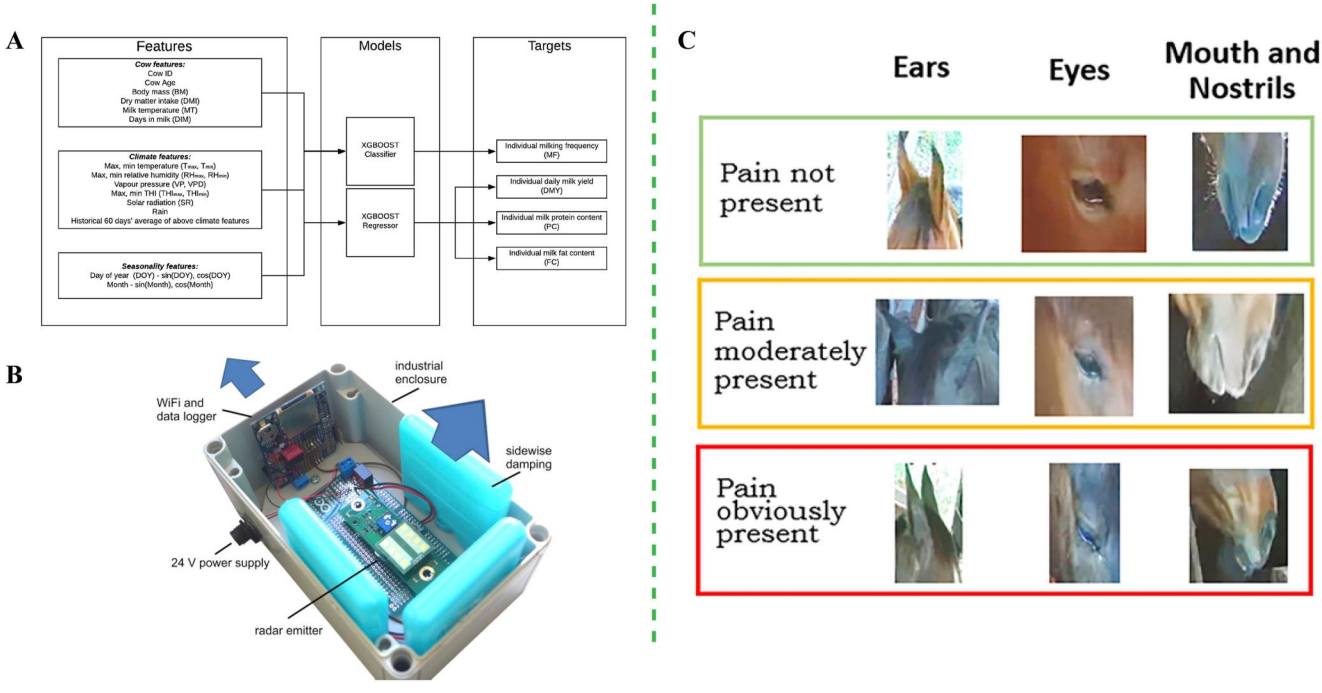


FIGURE 2 The impact of AI technology on animal behavior and emotion recognition. (A) This machine learning framework is integrated with a robotic milking system to predict the milk production, milking frequency, and milk quality of each cow, thus determining healthier milking strategies for cows [11]. (B) By collecting daily activity variations of sows through radar sensors, it enables monitoring of increased sow activity and prediction of sow parturition [12]. (C) The Horse Grimace Scale monitors pain responses in horses by analyzing their ear, eye, and muzzle features [16].

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labor-intensive, less accurate, and prone to repetitive counting and omissions. In a study by K. Jiang et al., they addressed this issue by incorporating an attention mechanism into the deep learning algorithm YOLOv7, resulting in an improved version called CBAM-YOLOv7 [14]. The experimental results demonstrate that CBAM-YOLOv7 achieves higher accuracy and slightly improves the recall rates, mAP@0.5 and mAP@0.5:0.95, compared to SE-YOLOv7 and ECA-YOLOv7, respectively. The evaluation metrics of CBAM-YOLOv7 show greater improvement. This indicates that the proposed intelligent duck counting method in this study is feasible and can contribute to the development of intelligent and reliable automated duck counting. This model enables real-time monitoring of the population density of Muscovy ducks in dense groups, thereby promoting the development of the smart farming industry.

Animal emotion recognition

AI applied to animal emotion recognition can provide valuable insights and methods for animal behavior research and welfare [43]. By utilizing advanced image processing and machine learning techniques, we can accurately analyze and identify the emotional states of animals [5]. This application helps us better understand their emotional experiences and needs, thereby improving their quality of life and welfare [44]. Moreover, it can play a crucial role in animal conservation by helping monitor their health, identify potential disease risks, and take necessary intervention measures in a timely manner [45]. Through the application of AI in animal emotion recognition, we can comprehensively focus on and protect the emotional needs of animals [46].

Research on determining the emotional states of animals has been a topic of great interest among animal welfare researchers [47]. One study suggests that the internal emotional states of domestic cats can be captured through facial morphological changes associated with pain [15]. By employing a machine learning model using CNNs (ResNet50) and geometric landmark analysis based on a specific species' facial action coding system (i.e., catFACS), researchers analyzed facial images of domestic shorthair cats at different time points corresponding to varying levels of pain during ovariohysterectomy. The automatic recognition of pain achieved an accuracy rate of over 72%, indicating the potential application of this intelligent recognition strategy in animal emotion recognition.

Dogs, as significant companion animals, typically require training to adapt to various environments. During the training process, the emotional state of the canines plays a crucial role in determining the effectiveness of the training [48–50]. Excessive punitive training not only leads to decreased canine morale but also severely compromises their animal welfare [51, 52]. To address this issue, the present study designed an emotion recognition system for dogs [53]. Specifically, they initially trained this model using 13,809 annotated images of canines. By observing limb movements such as the angle between the tail and spine, the height of the head lift, and the curvature of the limbs, they were able to determine the emotional state of the dogs. Subsequently, they validated the model using 400 full-body images of dogs representing four emotions: anger, fear, happiness, and relaxation. The research findings demonstrated that this emotion recognition approach could intuitively identify different emotional states in dogs, achieving an accuracy rate ranging from 60% to 70%. This effectively safeguards the animal welfare of canines during training [53].

Visual algorithms have shown potential not only in pets but also in assessing the degree of pain in horses. Lencioni et al. developed a visual algorithm based on the automatic calculation classifier of the Horse Grimace Scale (HGS) (Figure 2C) [16]. By training the algorithm using machine learning techniques, they were able to monitor the pain responses of horses more accurately and in real time. This provides earlier diagnosis and more effective treatment for affected animals. When categorizing pain into three levels: absent, moderate, and obvious, the overall accuracy was 75.8%. When classifying between two categories, pain absent and pain present, the overall accuracy reached 88.3%.

AI TECHNOLOGY ENSURING ANIMAL NUTRITION AND HEALTH

AI revolution in animal nutrition

AI has revolutionized animal nutrition by enabling practical applications that have transformed the management and optimization of animal diets [17]. One notable application involves the development of AI-driven feeding systems. These systems utilize AI algorithms to accurately compute and dispense precise quantities of feed tailored to the specific nutritional requirements of individual animals [54]. By analyzing data such as animal weight, body condition, and growth rate, these algorithms generate customized feeding plans [55]. These plans not only promote optimal growth and health but also reduce waste and mitigate the risks associated with overfeeding or underfeeding [56].

The accurate collection of feed intake data is crucial in the creation of an automated feeding system [57]. Analyzing this data enables meticulous statistical analysis, which is currently lacking in most commercial dairy farms [58]. One recommended approach suggests using visual algorithms in conjunction with RGBD cameras. These cameras capture red, green, blue, and

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depth information to impeccably gather individual feed intake data [17]. The author has developed a transfer learning (TL) model based on CNNs. This model incorporates a CNN model trained for two feed types, a multilayer perceptron, and a CNN model trained for two feed types along with classification data (Figure 3A). These models collaborate with the RGBD camera to analyze a series of images capturing the weight of diverse feed types collected under diverse configurations and lighting conditions within a research-based dairy farm. Rigorous experimentation has demonstrated the effectiveness of the TL model, resulting in a maximum error of 0.14 kg per meal and a root mean square error of 0.19 kg [17]. This model holds great potential for accurately quantifying individual feed intake in barn-kept cows, providing invaluable guidance for tailored feeding plans.

With the rapid development of artificial intelligence, Internet of Things (IoT), and digital information processing technologies, pet feeding systems are no longer limited to simple adjustments in food quantity [59, 60]. Nowadays, an increasing number of pet feeding devices focus on the animal welfare of companion animals like cats and dogs. One study developed an automatic model for estimating pet food amounts [18]. This model analyzed and compared different machine learning algorithms used in pet care applications by utilizing data-driven intelligent feeding systems. It also created a virtual database incorporating various variables such as the pet's growth, temperature, heart rate, eating habits, activity, sleep, and urine pH level. Researchers trained and validated supervised machine learning models, including linear regression, Gaussian process regression, narrow neural networks, linear support vector machines, and fine trees, using this database. A comparison of different models revealed that the interactive linear regression model exhibited the best efficiency in analyzing animal welfare-related data. Further integration of this model into hardware and the IoT can lead to the development of smart feeding products that address issues such as pet obesity, thus safeguarding the welfare and safety of companion animals [18].

Advancing animal health through intelligent systems

AI is revolutionizing our comprehension of animal welfare and playing a crucial role in monitoring and maintaining of animal health [61]. Noninvasive AI-based detection techniques curtail human intervention while proficiently and precisely amassing data, thus establishing a novel benchmark for augmenting animal welfare [62].

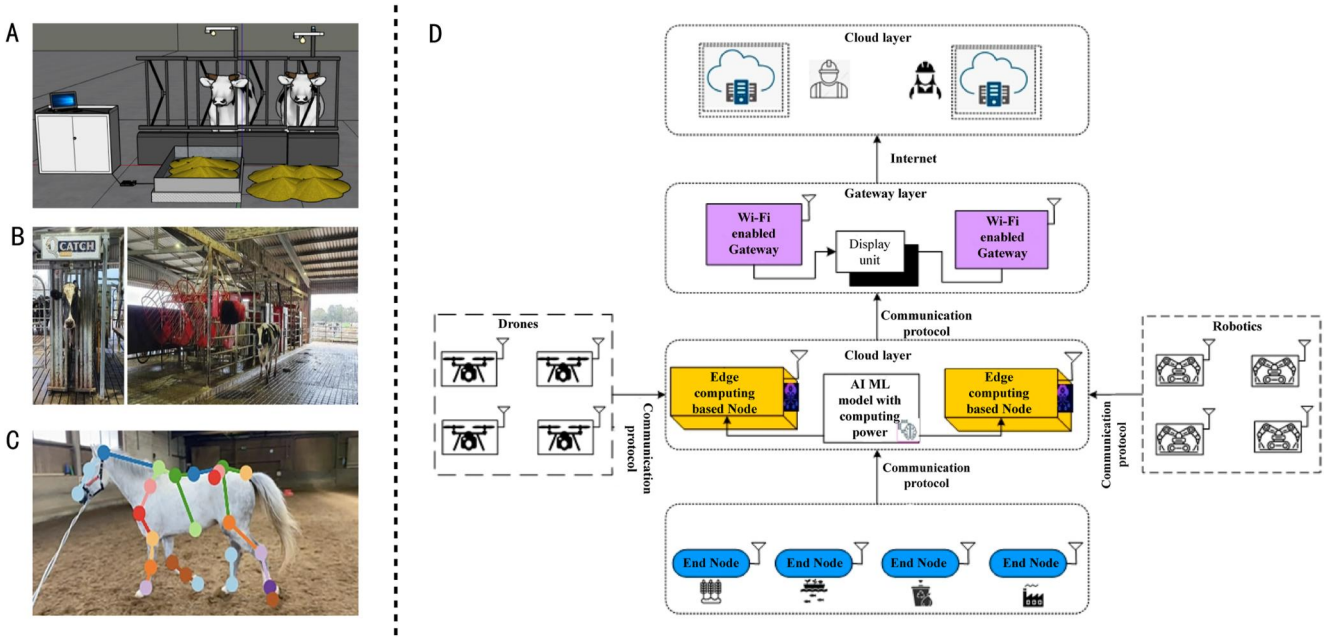


FIGURE 3 AI technology ensuring animal nutrition and health. (A) A transfer learning model based on convolutional neural networks is used to determine the feeding behavior of cows using images from within the dairy farm [17]. (B) A computer vision approach utilizing images and infrared thermal videos is employed to predict the eye temperature of cows, milk yield, and milk quality, enabling health monitoring of cows [19]. (C) AI interprets the limb movements of horses through visual images to distinguish between healthy and lame horses [20]. (D) An intelligent ecosystem combining technologies such as smart systems, Internet of Things, and blockchain is implemented to promote the dairy industry and improve animal welfare [9].

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In the field of animal welfare, noninvasive methods assume paramount importance for their capacity to minimize human intrusion while efficiently and accurately acquiring data. Fuentes et al. have pioneered a noninvasive computer vision approach employing RGB cameras and infrared thermal videos (IRTV) for data acquisition (Figure 3B) [19]. They have also employed machine learning models to predict eye temperature, milk yield, and quality. In their study involving 102 dairy cows, they demonstrated the model's remarkable accuracy (R = 0.96) and its potential for application in both animal welfare and the modeling of biological and non-biological pressures.

Feuser et al. have developed a diagnostic software tool that leverages AI technology to promptly identify lameness in horses and mitigate its impact on their well-being (Figure 3C) [20]. This software can differentiate between lame and healthy horses while identifying the affected limbs. By analyzing 22 horses with unilateral forelimb or hindlimb lameness and comparing them to a control group of eight healthy horses, they have developed a program based on pose estimation capable of discerning different types of lameness and healthy horses. This technology holds promise for studying limb and body movement in both human and veterinary sciences without requiring any attachments to the subjects.

Gehlot et al. have explored the potential of AI technology to significantly enhance animal welfare in dairy cows (Figure 3D) [9]. They have discussed intelligent systems that incorporate sensory, monitoring, and analytical techniques for diverse applications, including animal health monitoring, animal tracking, milk quality and supply chain, feed monitoring, and safety. Furthermore, they have proposed a flexible architecture adaptable to research and development advancements in data collection, in addition to applications involving edge devices, robots, drones, and blockchain, thus establishing intelligent ecosystems [9]. Their developed system, bolstered by technologies like the IoT, facilitates the creation of a smart ecosystem for dairy farming, making noteworthy contributions to improving animal welfare in the dairy industry.

The rapid economic development in China has led to a significant growth in the pet market [63]. However, challenges remain regarding welfare standards and care strategies for companion animals in China [64]. There is an urgent need for improved products and policies to enhance conditions for pets. In response to this, Xu et al. proposed a design for an intelligent collar and accompanying facilities [65]. This design integrates the collar with Alibaba Cloud services platforms and the MQTT IoT access protocol to enable real-time collection of pet activity levels, heart rate, and location data. The collar incorporates sensors such as a JY901 gyroscope, MAX30102 heart rate sensor, and AntrackerA9G positioning system to gather this information. Additionally, the design utilizes gravity sensors to automate feeding functions, monitor weight, and ensure regular removal of pet waste [65].

The overall health status of companion animals is a top concern for pet owners. However, most people lack knowledge about pet-related diseases beyond professionals [66]. As a result, the behaviors of many companion animals often hint at potential diseases or fragile health conditions, but owners are unable to take timely and effective measures [67]. To address this issue, a study proposed an integrated framework for an intelligent pet care system to monitor the health conditions of household pets [68]. The framework utilizes Internet of Medical Things technologies and remote monitoring devices such as cameras and electrocardiogram sensors to gather health information about the pets. This information is then transmitted through the FogBus fog computing platform and analyzed using a time series artificial neural network predictive model to measure real-time health vulnerability probabilities (PoHV) and time-sensitive measurements (TSM) for household pets. PoHV is used to assess the health status index of animals, while TSM provides analytical results on the development and health conditions of animal diseases over time and space. Finally, the data are visualized and presented to caregivers and pet owners using self-organizing mapping technology [68]. The system has been experimentally validated on American bulldogs, achieving high accuracy, precision, sensitivity, and f1-measure values all above 93% for disease monitoring.

The development of noninvasive video-based biometric systems and the application of machine learning techniques have opened up possibilities for comprehensive animal health monitoring and automated welfare assessment [69]. Recent studies have mentioned the extraction of data from cows through noninvasive video collection and the use of biostatistical analysis techniques to develop two machine learning models [21]. The first model focuses on biostatistical regression for parameters such as somatic cell count, body weight, rumination, and feed intake. The second model utilizes facial feature mapping to predict the age of the animals. These models were tested in a robotic dairy farm, and the results showed high accuracy and low error rates, indicating potential applications in evaluating milk productivity, quality traits, and welfare in dairy farms.

THE ROLE OF AI IN REVOLUTIONIZING ANIMAL LIVING ENVIRONMENTS

The revolutionary application of AI in animal living environment management

AI has revolutionized livestock and poultry management, exerting a significant influence on animal health

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and productivity. Environmental parameters, such as temperature, humidity, and air quality, play crucial roles in these settings [70]. AI-driven environmental monitoring systems utilize sensors and data analytics to continuously assess these conditions [71]. When deviations from the optimal range occur, AI systems can automatically adjust ventilation, heating, or cooling parameters to create a more comfortable and healthy environment for the animals [13]. This not only enhances animal welfare but also improves production outcomes by reducing stress-related issues and disease prevalence.

Previously, environmental monitoring in animal farming mainly focused on toxic gases like ammonia [72]. However, in response to growing concerns about animal welfare, monitoring methodologies for gases such as carbon dioxide and sulfur dioxide have emerged. These approaches aim to prevent livestock from inhabiting unhealthy environments. A research study introduces a method for constructing a model capable of predicting four distinct gases: carbon dioxide, sulfur dioxide, nitrogen dioxide, and atmospheric particulate matter [22]. The study combines linear regression, support vector regression, and gradient boosting decision tree (GBDT) ensemble models to formulate an air quality index prediction model. Various sensors designed for air quality analysis are also integrated. Models with lower errors were selected based on performance evaluations. By incorporating algorithms from AI systems and integrating observational data to rectify inaccuracies, real-time forecasting of the levels of five air pollutants (carbon dioxide, sulfur dioxide, nitrogen dioxide, PM2.5, and PM10) can be achieved. This advancement presents opportunities for enhancing the living conditions of animals.

Li and his team have designed an intelligent monitoring system specifically for stacked chicken coops [23]. The sensor distribution topology, tailored to the complex structure of the henhouse, enables real-time monitoring of temperature, CO_2 levels, and wind speed in different layers of the chicken coop. The data are analyzed and organized on a cloud server. Through a smartphone app, users can instantly check the environmental conditions of the chicken coop. This system enables data analysis and transmission from sensors to monitoring terminals to smart mobile devices, allowing remote monitoring of the chicken coop environment. It holds significant practical significance for precision breeding in large-scale livestock farming.

Sun et al. have developed an automated monitoring and control system for livestock and poultry housing environments using IoT and robotics technology (Figure 4A) [24]. This system utilizes IoT robots equipped with sensors to collect environmental data within the animal housing. It leverages WiFi, LoRa wireless communication, and IoT technology to control the actuators inside the housing, enabling automatic monitoring and adjustment of the livestock and poultry farming environment. The system management platform is built using Java and C# programming languages, providing comprehensive management and control of multiple zones within the housing. The authors aim to ensure animal welfare throughout the farming process while simultaneously maintaining the quality of animal growth and economic benefits.

The application of automated cleaning and disinfection systems in animal facilities

Automated cleaning and disinfection systems, often utilizing robotic technology, have become essential for maintaining sanitary conditions in livestock and poultry facilities [73]. AI-driven robots, equipped with sensors and cameras, efficiently navigate the environment and identify areas requiring cleaning and disinfection. These robots effectively remove organic matter, such as manure, and accurately apply disinfectants as necessary. Automating this critical aspect of animal housing not only reduces the workload of farm personnel but also ensures a consistently clean and pathogen-free environment [25]. Consequently, it contributes to disease prevention, improves animal health, and enhances overall hygiene standards.

To address the challenge of disinfection in livestock farms, a study aims to develop an intelligent strategy by creating smart robots equipped with disinfection sprayers (Figure 4B) [25]. This robot consists of an automated vehicle, a disinfectant spraying unit, a monitoring unit, and a controller unit. It supports both automatic and remote operation and can follow designated routes. The use of this automated smart robot effectively resolves the issues associated with slow and ineffective manual disinfection, thereby upholding overall hygiene standards for livestock.

Another study adopts a similar approach by employing a robot guided by magnetic-RFID navigation (Figure 4C) [26]. This robot comprises a mobile platform, a disinfection spraying device, environmental monitoring sensors, and a controller. Additionally, it is equipped with air-assisted nozzles capable of independently atomizing and dispersing the liquid disinfectant. Actual performance tests conducted in poultry houses demonstrated that the robot autonomously navigates at speeds ranging from 0.1 to 0.5 m/s, with a maximum deviation of 50.8 mm from the expected trajectory. The air-assisted nozzles enable simultaneous atomization and diffusion of the liquid disinfectant, making it suitable for spraying liquid at flow rates of 200–400 ml/min. The resulting droplet diameter (DV.9) varies from 51.82 to 137.23 μL, increasing with the liquid flow rate. The

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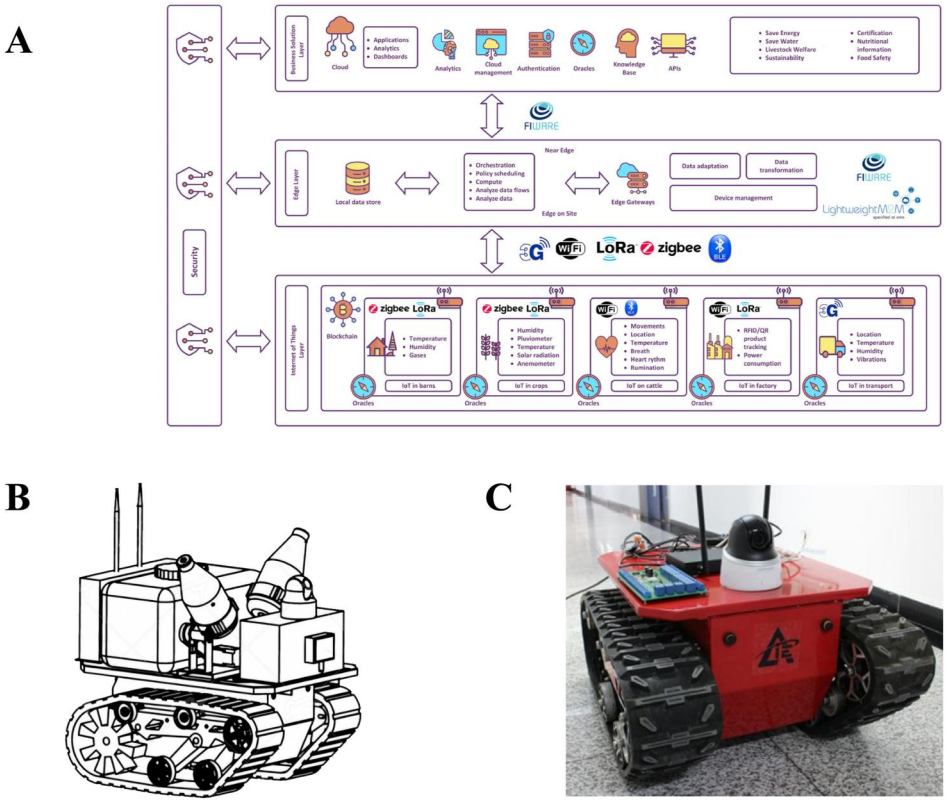


FIGURE 4 The role of AI in revolutionizing animal living environments. (A) Intelligent robots for automatic detection and control of livestock farming environments [24]. (B) Remote-controllable intelligent robots for automatic disinfection [25]. (C) Disinfection robots guided by magnetic radio-frequency identification navigation [26].

deposition density of the spray droplets generated by the nozzles ranges from 116 to 149/cm^2, decreasing with the spray distance. These parameters of liquid droplet size and density meet the necessary criteria for effectively eliminating adherent pathogenic microorganisms in different areas of the cages.

ADVANCEMENTS IN AI-DRIVEN ANIMAL DISEASE PREDICTION, MONITORING, AND DIAGNOSIS

Animal disease prediction and monitoring

AI technology has brought about a revolution in the field of animal disease prediction and monitoring. By utilizing advanced machine learning algorithms, AI systems can analyze extensive datasets, including animal health records, environmental factors, and genetic information, to identify patterns and accurately forecast disease outbreaks [74]. These AI-powered solutions enable the early detection of diseases, facilitating timely intervention and preventing the spread of infections [75, 76]. Moreover, AI algorithms continuously monitor animal health parameters in real time, providing valuable insights for farmers and veterinarians to optimize disease prevention strategies and improve animal welfare [77]. Overall, AI technology plays a crucial role in safeguarding animal health, enhancing disease management practices, and ensuring sustainable agriculture.

In a recent study, researchers proposed an intelligent framework for real-time monitoring of animal diseases, using swine flu as an example [27]. They employed a fog-centric, IoT-based intelligent health care support system to monitor and control the outbreak of the swine flu virus. The researchers used the iFogSim simulator to evaluate various parameters such as accuracy, energy consumption, and latency by simulating IoT devices and fog nodes. Additionally, they utilized WEKA to develop a hybrid classifier. The results indicated that the integration of fog computing and cloud computing services yielded enhanced network bandwidth reliability, elevated operational efficiency, and reduced response times in comparison to conventional cloud computing models. This integration facilitated real-time notifications.

In another study, a fog computing-assisted end-to-end animal behavior analysis and disease monitoring platform called SmartHerd was specifically designed for dairy farms (Figure 5) [28]. The platform relies on wearable sensors attached to the cows' bodies for data

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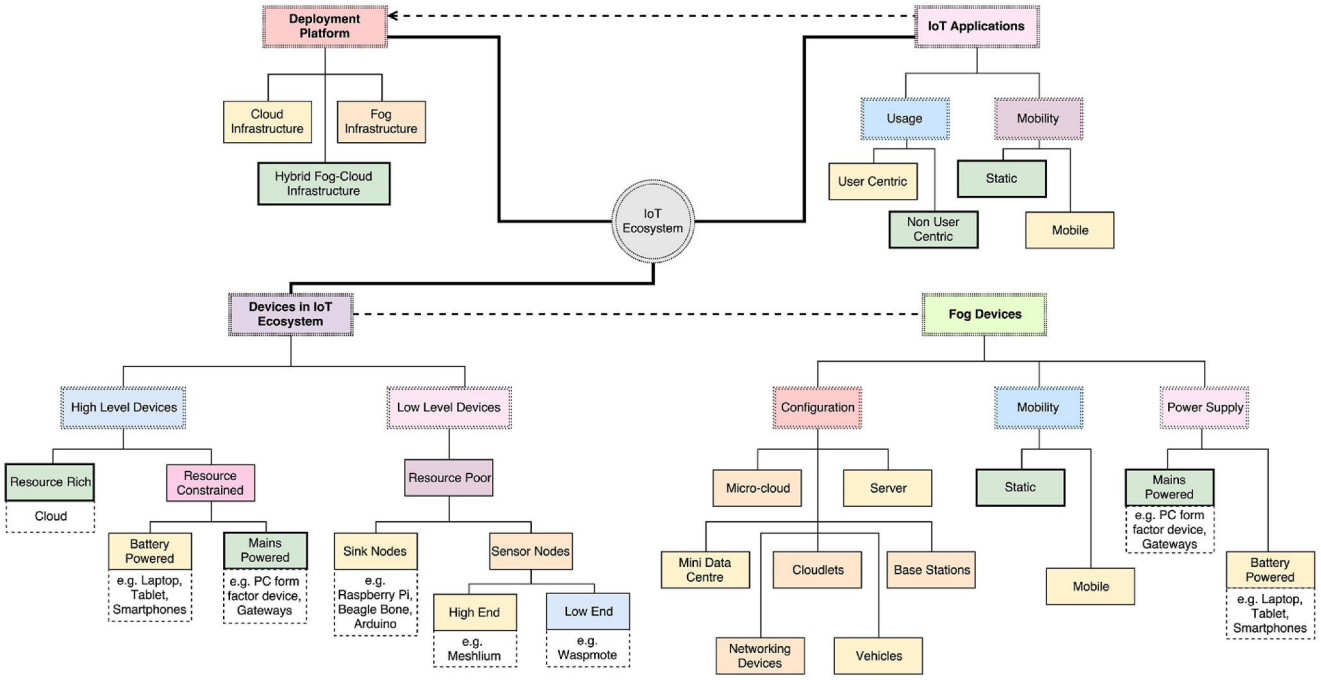


FIGURE 5 A fog computing assisted for animal behavior analysis and disease monitoring platform [28].

collection. Through data classification and analysis on the platform, the authors aimed to provide farm owners with valuable insights. To address the main issue of limited Internet connectivity in remote farms, they proposed adopting a microservice-oriented design and distributed computing paradigm.

Chatterjee et al. proposed a framework known as LiveCare, based on the IoT, for automated monitoring of the health status of cows in large-scale dairy farms [29]. This framework tracks the behavioral changes of cows on a daily basis and introduces the Cow Disease Prediction (CDP) algorithm as a core component. The CDP algorithm analyzes the behavioral changes of cows to predict various cow diseases.

Feng et al. introduced a framework based on the IoT for sensing animal social behavior [30]. This framework was applied to simulate the spread of mastitis in dairy cows and infer the risk of cows being infected with mastitis. By deploying portable GPS devices on cows and tracking their movement patterns and interactions, a directed and weighted social behavior graph of cows was established. The researchers proposed a disease transmission probability model that considered both direct contact with infected cows and indirect contact through environmental contamination. This model was used to estimate and predict the probability of mastitis infection and help identify infected cows in the dairy farm, giving them priority for further screening. The effectiveness of this framework was demonstrated through experiments, and the results aligned with somatic cell count (SCC) mastitis tests.

Diagnosis of animal diseases

AI technology has emerged as a powerful tool in diagnosing animal diseases, utilizing comprehensive data analysis of clinical symptoms, laboratory results, and medical imaging [78]. By identifying patterns, AI algorithms can accurately diagnose diseases, leading to a significant reduction in diagnostic errors and improved efficiency in veterinary practices [79]. Moreover, AI-based diagnosis enables early disease detection, facilitating prompt treatment and better outcomes for animals [32]. Continuous learning and improvement in AI technology hold immense promise for revolutionizing animal disease diagnosis, benefiting both animals and veterinary professionals [80].

Blömke and their team designed an automated system to assess pig welfare indicators and diagnose ear and tail lesions in pigs at slaughterhouses [31]. The system underwent testing on a sample of 5952 pigs and was compared with assessments made by skilled veterinarians. The results demonstrated that the system exhibited high sensitivity (above 77%), specificity (above 96%), and accuracy (above 95%) in detecting lesions in both the ears and tails. This finding is significant for diagnosing pig ear and tail diseases and ensuring the welfare of these animals.

AI has proven to be an effective solution for addressing challenges faced by remote farms with limited diagnostic capabilities. In a recent study, researchers proposed the use of a convolutional neural network (CNN) to enhance the diagnostic accuracy of

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an Animal Disease Diagnosis Expert System (ADDES) for swift and precise identification of animal diseases [32]. The study utilized a dataset of 1200 images representing 15 distinct diseases. By extracting features from these images and employing the CNN for pattern recognition, the researchers achieved an impressive accuracy rate of 97.06%. The autonomous disease diagnosis system developed through this approach reduces reliance on veterinarians, making it especially valuable for farms lacking veterinary expertise or facing challenges in maintaining optimal animal husbandry practices. By harnessing the power of AI and its ability to analyze extensive amounts of data, including clinical symptoms, laboratory results, and medical imaging, AI algorithms can identify patterns and provide accurate diagnoses. These AI-powered systems have the potential to significantly reduce diagnostic errors and enhance the efficiency of veterinary practices. Moreover, AI-based diagnosis enables early disease detection, allowing for prompt treatment and improved outcomes for animals. Through continuous learning and improvement, AI technology holds immense promise for revolutionizing the field of animal disease diagnosis, ultimately benefiting both animals and veterinary professionals [32].

Heat stress has detrimental effects on the health and productivity of cows. Most current intelligent models for heat stress assessment rely on predetermined relationships between environmental conditions and cows' physiological responses, established by researchers [81, 82]. To overcome potential biases caused by human judgment, a study employed machine learning algorithms to evaluate the impact of environmental heat stress factors (air temperature, relative humidity, solar radiation, and wind speed) which affect cows' physiological responses (respiratory rate, skin temperature, and vaginal temperature) [33]. Machine learning algorithms automatically consider the nonlinearity of the data, eliminating subjectivity in the selection of predictor–response relationships. The study developed computational models using penalized linear regression, random forest, gradient boosting machine, and neural networks, which can effectively help farmers take evidence-based interventions before expected stressful environmental conditions occur [33]. Early interventions can improve animal health and provide effective support for ensuring animal welfare.

Visual algorithms not only enable individual recognition but also find relevant applications in heat stress assessment. S. Fuentes et al. developed an AI-based noncontact biometric system for automated animal welfare assessment in farm and transportation processes [3]. The system utilizes an automated tracking algorithm to extract skin temperature as a head feature from IRTVs. Two parameter engineering procedures were employed to extract the heart rate (HR) in beats per minute (BPM) and the respiratory rate (RR) from RGB videos. The overall fitting rate reached 96%, making it an effective tool for heat stress assessment in live sheep.

DISCUSSION AND FUTURE PROSPECTS

AI offers significant advantages for enhancing animal welfare [61]. Primarily, AI enables meticulous recognition of animal behaviors, emotions, as well as the prognosis and diagnosis of diseases. This level of accuracy allows for timely interventions, augmenting the overall welfare of animals. Additionally, AI systems automate tasks including environmental management and facility sanitation, minimizing human errors, and ensuring uniform quality in animal care [22, 27]. Furthermore, AI revolutionizes animal nutrition and health, formulating individualized nutrition strategies and proactively addressing health concerns before they escalate.

However, the application of different AI techniques also has certain limitations (Figure 6). For example, computer vision-based methods like CNNs can accurately recognize animals and their behaviors, but may struggle in low light or cluttered environments [83, 84]. Sensor-based approaches efficiently collect behavioral data, yet interpreting complex emotions remains difficult [85]. ML techniques effectively analyze nutrition and health parameters, but lack human-like comprehension of animal needs [86]. Knowledge-based expert systems enable disease diagnosis, but rely on predefined rules and have limited learning capabilities. Overall, a combination of different AI methods can help overcome individual limitations and provide more robust solutions.

To develop and train AI systems, a substantial amount of data are essential [87]. However, in certain cases, data availability may be limited or difficult to obtain, posing challenges to the effectiveness of AI applications [88, 89]. While AI can effectively monitor the physiological indicators of animals, understanding their emotions and behaviors remains beyond its capabilities. Consequently, there is a risk of failing to identify crucial welfare issues. Additionally, the successful implementation of AI technology requires ongoing training and maintenance, necessitating personnel with the relevant knowledge and skills to manage and utilize these systems. Unfortunately, the absence of such knowledge and training can potentially impede the adoption of this technology. Implementation of AI systems, especially in expansive facilities, can incur significant initial expenses, potentially hindering adoption by smaller entities or individual proprietors [90]. Moreover, persistent surveillance of animals via AI ignites ethical debates concerning their privacy and autonomy [91]. A judicious equilibrium between welfare monitoring and preserving their independence is vital.

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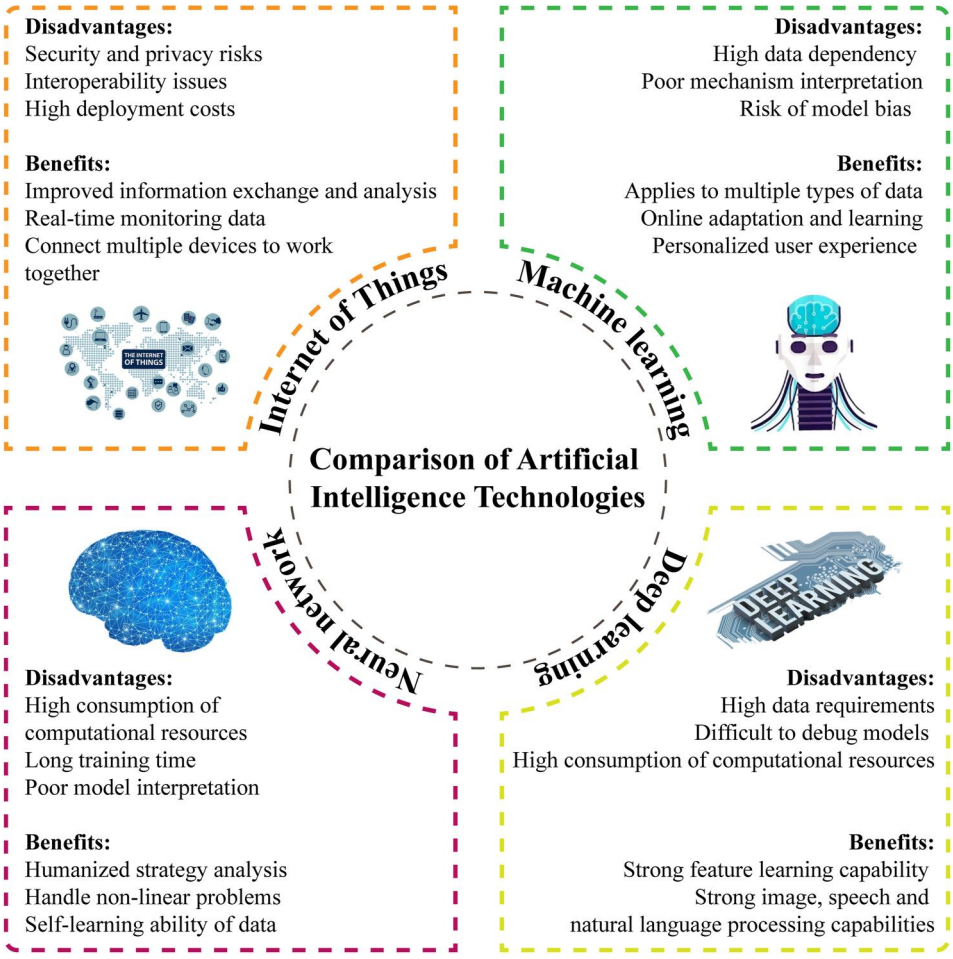


FIGURE 6 The Summary of the Advantages and Disadvantages of Several Technologies. Icons were from www.freecodecamp.org/news/introduction-to-iot-internet-of-things/, www.futura-sciences.com/tech/definitions/intelligence-artificielle-deep-learning-17262/, datascience.eu/machine-learning/interpreting-your-deep-learning-model-by-shap/, www.mime.asia/the-difference-between-artificial-intelligence-deep-learning-and-machine-learning/.

To propel animal care advancements through AI, various avenues warrant exploration. Integration of AI with the IoT can foster more intelligent animal care facilities, facilitating perpetual monitoring and enhancement of all facets, from diet to habitat [4, 19]. Real-time data from IoT devices can be analyzed by AI algorithms to provide optimal care for animals [9]. Secondly, the future may witness the development of AI systems capable of personalizing care plans for individual animals based on their unique behaviors, emotions, and health needs [16, 46]. Similar to how AI customizes ads for individuals, personalized animal care can lead to better outcomes. Finally, collaborative research involving technologists, veterinarians, and animal behaviorists is essential. This interdisciplinary approach ensures that AI technology in animal care is developed and implemented in a way that truly benefits the animals. By combining expertise from different fields, advancements can be made in AI algorithms, data collection, and ethical considerations, ultimately improving animal welfare.

By leveraging the precision, automation, and holistic management capabilities of AI, addressing the limitations and concerns, and exploring future directions, the potential for enhancing animal welfare through AI technology is promising. With continued research and responsible implementation, the integration of AI in animal care will likely contribute to the well-being of animals, sustainable agriculture, and veterinary practices.

CONCLUSION

In conclusion, the utilization of AI offers substantial potential for enhancing animal welfare. By capitalizing on its advantages, addressing limitations and concerns, and exploring future directions, we can make significant progress in this field. To achieve this, it is crucial to continue conducting research, implement AI technology responsibly, and foster interdisciplinary collaboration. These efforts will contribute to improved animal well-

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being, sustainable agriculture, and advancements in veterinary practices. However, it is vital that we remain mindful of ethical considerations and ensure that AI is used in a responsible and ethical manner, thereby creating a brighter future for animals.

AUTHOR CONTRIBUTIONS

Conceptualization, methodology and writing—original draft preparation, Li Zhang; writing—review and editing, Li Zhang, Wenqiang Guo, Chenrui Lv, Meng Guo., Mei Yang, Qiuyue Fu and Xiaomeng Liu; supervision, Li Zhang; funding acquisition, Li Zhang All authors have read and agreed to the published version of the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

No new data were created or analyzed in this study.

ETHICS STATEMENT

This research involved no experimental animal or human participant and did not contain offensive language.

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